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R LAB OUTPUTS:

1.1:

```
> roll_no <- 102# Integer
> name <- "harshi"# Character
> cgpa <- 8.35# Numeric
> pass <- TRUE# Logical
> roll_no
[1] 102
> name
[1] "harshi"
> cgpa
[1] 8.35
> pass
[1] TRUE
```

1.2:

```
> emp_id <- 3001
> salary <- 35000.75
> department <- "finance"
> is_permanent <- TRUE
> emp_id
[1] 3001
> salary
[1] 35000.75
> department
[1] "finance"
> is_permanent
[1] TRUE
```

1.3:

```
> product_name <- "mobile"
> price <- 50000
> stock_available<-product_name <- "mobile"
> price <- 50
> discount_eligible <- FALSE
> product_name
[1] "mobile"
> price
[1] 50
> discount_eligible
[1] FALSE
> stock_available
[1] "mobile"
```

1.4:

```
> city <- "Hyderabad"
> temperature <- 32.5
> rain <- FALSE
> wind_speed <- 10.8
> city
[1] "Hyderabad"
> temperature
[1] 32.5
> rain
[1] FALSE
> wind_speed
[1] 10.8
>
```

1.5:

```
> account_no <- 987654321
> holder_name <- " Kumar"
> balance <- 23000.50
> account_actve <- TRUE
> account_no
[1] 987654321
> holder_name
[1] " Kumar"
> balance
[1] 23000.5
> account_actve
[1] TRUE
```

1.6:

```
> subject <- "data science"
> total_marks <- 50
> marks_obtained <- 40
> result <- marks_obtained >= 20
> total_marks
[1] 50
> marks_obtained
[1] 40
> result
[1] TRUE
```

1.7:

```
> movie_title <- "thammudu"
> ratng <- 8.00
> reviews <- 1200
> available <- TRUE
> movie_title
[1] "thammudu"
> ratng
[1] 8
> reviews
[1] 1200
> available
[1] TRUE
```

1.8:

```
> vehicle_count <- 300
> locaton <- "Highway Road"
> avg_speed <- 55.6
> congested <- TRUE
> vehicle_count
[1] 300
> locaton
[1] "Highway Road"
> avg_speed
[1] 55.6
> congested
[1] TRUE
>
```

2.1:

```
> age <- 27
> income <- 38000
> if(age > 21 & income > 30000){
+   print("Eligible for Loan")
+ } else {
+   print("Not Eligible for Loan")
+ }
[1] "Eligible for Loan"
```

2.2:

```
> attendance<-80
> if(attendance >= 85){
+   print("Eligible for Exam")
+ } else {
+   print("Not Eligible for Exam")}
[1] "Not Eligible for Exam"
```

2.3:

```
> amount <- 6200
> premium_member <- TRUE
> if(amount > 10000 | premium_member){
+   print("Discount Applied")
+ } else {
+   print("No Discount")
+ }
[1] "Discount Applied"
```

2.4:

```
> units <- 350
> if(units < 100){
+   print("Low Usage")
+ } else if(units <= 300){
+   print("Medium Usage")
+ } else {
+   print("High Usage")
+ }
[1] "High Usage"
```

2.5:

```
> experience <-10
> ratng <- 5
> if(experience > 5 & ratng > 4){
+   print("Bonus Eligible")
+ } else {
+   print("Not Eligible for Bonus")}
[1] "Bonus Eligible"
```

2.6:

```
> temperature <- 50
> if(temperature < 0 | temperature > 40){
+   print("Extreme Temperature Warning")
+ } else {
+   print("Normal Temperature")
+ }
[1] "Extreme Temperature Warning"
```

2.7:

```
> marks <- 85
> if(marks >= 90){
+   grade <- "A"
+ } else if(marks >= 75){
+   grade <- "B"
+ } else if(marks >= 50){
+   grade <- "C"
+ } else {
+   grade <- "Fail"
+ }
> print(grade)
[1] "B"
```

2.8:

```
> age <- 40
> income <- 50000
> credit_score <- 720
> if(age >= 21 & age <= 60 & income >= 25000 & credit_score >= 700){
+   print("Credit Card Approved")
+ } else {
+   print("Credit Card Rejected")
+ }
[1] "Credit Card Approved"
```

2.9:

```
> bp <- 150
> sugar <- 200
> if(bp > 140 & sugar > 200){
+   risk <- "High Risk"
+ } else if((bp >= 120 & bp <= 140) | (sugar >= 140 & sugar <= 200)){
+   risk <- "Medium Risk"
+ } else {
+   risk <- "Low Risk"
+ }
> print(risk)
[1] "Medium Risk"
```

2.10:

```
[1] "Medium Risk"
> vehicle_count <- 190
> if(vehicle_count > 200){
+   print("Long Green Signal")
+ } else if(vehicle_count >= 100){
+   print("Medium Green Signal")
+ } else {
+   print("Short Green Signal")
+ }
[1] "Medium Green Signal"
```

2.11:

```
> atendance <- 95
> performance <- 8.6
> project_completed <- TRUE
> if(atendance >= 90 & performance >= 9& project_completed){
+   print("Eligible for Appraisal")
+ } else {
+   print("Not Eligible for Appraisal")
+ }
[1] "Not Eligible for Appraisal"
```

2.12:

```
> face_detected <- TRUE
> multple_faces <- FALSE
> internet_connected <- TRUE
> if(!face_detected | multple_faces | !internet_connected){
+   print("Exam Disqualified")
+ } else {
+   print("Exam Allowed")
+ }
[1] "Exam Allowed"
```

2.13:

```
> high_demand <- TRUE
> available_drivers <- 40
> if(high_demand & available_drivers < 50){
+   print("Surge Pricing Applied")
+ } else {
+   print("Normal Pricing")
+ }
[1] "Surge Pricing Applied"
```

2.14:

```
> temperature <- 75
> pressure <- 130
> vibration <- 6
> if(temperature > 80 | pressure > 120 | vibration > 7){
+   print("Machine Shutdown")
+ } else {
+   print("Machine Running Normally")
+ }
[1] "Machine Shutdown"
```

2.15:

```
> entrance_score <- 70
> twelfth_marks <- 90
> age <- 20
> if(entrance_score >= 70 & twelfth_marks >= 75 & age <= 25){
+   print("Admission Granted")
+ } else {
+   print("Admission Rejected")
+ }
[1] "Admission Granted"
```

3.1:

```
> simple_interest <- function(p, r, t){
+   si <- (p * r * t) / 100
+   return(si)
+ }
> simple_interest(10000, 6, 3)
[1] 1800
>
```

3.2:

```
> check_result <- function(marks){
+   if(marks >= 40){
+     return("Pass")
+   } else {
+     return("Fail")
+   }
+ }
> check_result(55)
[1] "Pass"
```

3.3:

```
> net_salary <- function(basic_salary){
+   tax <- basic_salary * 0.10
+   net <- basic_salary - tax
+   return(net)
+ }
> net_salary(35000)
[1] 31500
```

3.4:

```
>  electricity_bill <- function(units)
{
+     if(units <= 150){
+       bill <- units * 2
+     } else if(units <= 250){
+       bill <- (100 * 2) + (units - 1
00) * 3
+     } else {
+       bill <- (100 * 2) + (100 * 3)
+ (units - 200) * 5
+     }
+     return(bill)
+   }
> electricity_bill(550)
[1] 2250
```

3.5:

```
final_amount <- function(amount) {
  if (amount >= 5000) {
    discount <- amount * 0.20
  } else if (amount >= 2000) {
    discount <- amount * 0.10
  } else {
    discount <- 0
  }
  return(amount - discount)
}

# Call the function
final_amount(5500)
] 4400
```

3.6:

```
> celsius_to_fahrenheit <- function(c){
+   f <- (c * 9/5) + 32
+   return(f)
+ }
> celsius_to_fahrenheit(47)
[1] 116.6
```

3.7:

```
> assign_grade <- function(marks){
+   if(marks >= 90){
+     "A"
+   } else if(marks >= 75){
+     "B"
+   } else if(marks >= 50){
+     "C"
+   } else {
+     "Fail"
+   }
+ }
> assign_grade(78)
[1] "B"
```

3. 8:

```
> check_voting <- function(age){
+   if(age >= 18){
+     return("Eligible to Vote")
+   } else {
+     return("Not Eligible to Vote")
+   }
+ }
> check_voting(27)
[1] "Eligible to Vote"
```

3.9:

```
> calculate_emi <- function(principal, rate, years){
+   monthly_rate <- rate / (12 * 100)
+   months <- years * 12
+   emi <- (principal * monthly_rate * (1 + monthly_rate)^months) /
+     ((1 + monthly_rate)^months - 1)
+   return(round(emi, 2))
+ }
> calculate_emi(540000, 7.5, 10)
[1] 6409.9
>
```

3.10:

```
> hospital_bill <- function(consultation, room_charge, days)
+   subtotal <- consultation + (room_charge * days)
+   tax <- subtotal * 0.05
+   total <- subtotal + tax
+   return(total)
+ }
> hospital_bill(400, 3000, 4)
[1] 13020
```

3.11:

```
> order_total <- function(cart_value){
+   if(cart_value >= 3000){
+     discount <- cart_value * 0.10
+   } else {
+     discount <- 0
+   }
+   if(cart_value >= 5000){
+     delivery <- 0
+   } else {
+     delivery <- 100
+   }
+   final_amount <- cart_value - discount + delivery
+   return(final_amount)
+ }
> order_total(3200)
[1] 2980
>
```

3.12:

```
> calculate_gpa <- function(marks){
+   avg <- mean(marks)
+   if(avg >= 90){
+     result <- "Distinction"
+   } else if(avg >= 80){
+     result <- "First Class"
+   } else if(avg >= 50){
+     result <- "Second Class"
+   } else {
+     result <- "Fail"
+   }
+   return(result)
+ }
> calculate_gpa(c(88, 85, 90, 72, 80))
[1] "First Class"
>
```

3.13:

```
> smart_meter_bill<-function(peak_units,offpeak_units){
+   bill<-(peak_units*6)+(offpeak_units*3)
+   total_units<-peak_units+offpeak_units
+   if(total_units > 300){
+     bill <- bill * 1.10
+   }
+   return(bill)
+ }
> smart_meter_bill(190, 160)
[1] 1782
```

3.14:

```
> insurance_premium <- function(base, age, smoker){
+   premium <- base
+   if(age > 50){
+     premium <- premium * 1.20
+   }
+   if(smoker){
+     premium <- premium * 1.30
+   }
+   return(premium)
+ }
> insurance_premium(4000, 45, TRUE)
[1] 5200
```

3.15:

```
> payroll <- function(basic){
+   hra <- basic * 0.20
+   da <- basic * 0.10
+   gross <- basic + hra + da
+   tax <- gross * 0.12
+   net <- gross - tax
+   return(net)
+ }
> payroll(50000)
[1] 57200
```

3.16:

```
> water_usage_status <- function(litres){
+   if(litres <= 500){
+     "Normal"
+   } else if(litres <= 1000){
+     "High Usage"
+   } else {
+     "Critical"
+   }
+ }
> water_usage_status(1800)
[1] "Critical"
>
```

4.1:

```
+ )
> final_data <- rbind(student_data, new_students)
> final_data
  RollNo    Name Dept Marks
1    315  harshi  CSE    78
2    224   kalpa  ECE    85
3    265    ujju   IT    90
4    238 rithika  CSE    88
```

4.2:

```
> employees <- rbind(employees, new_employee)
> employees
  EmpID  Name    Dept Salary
1     1 priya      HR  35000
2     2 gowri      IT  45000
3     3  puji Finance  50000
4     4  Neha      IT  48000
```

4.3:

```
  PatientID    Name Age  Cost
1       201     ram  45 12000
2       202 Lakshmi  38 15000
> # Add new patient
> new_patient <- data.frame(
+   PatientID = 203,
+   Name = "henny",
+   Age = 50,
+   Cost = 18000
+ )
> patient_records <- rbind(patient_records, new_patient)
> patient_records
  PatientID    Name Age  Cost
1       201     ram  45 12000
2       202 Lakshmi  38 15000
3       203   henny  50 18000
>
```

4.4:

```
  ProductID ProductName Quantity
1         1         Pen      100
2         2        Book       50
3         3         Bag       30
> # Add new product
> new_product <- data.frame(
+   ProductID = 4,
+   ProductName = "scale",
+   Quantity = 75
+ )
> inventory <- rbind(inventory, new_product)
> inventory
  ProductID ProductName Quantity
1         1         Pen      100
2         2        Book       50
3         3         Bag       30
4         4       scale       75
```

4.5:

```
> final_records <- rbind(student_records, new_students)
> final_records
  RollNo   Name Dept Sem1 Sem2 Attendance
1      1   Arun  CSE   78   80         92
2      2 Tejesh  ECE   82   85         88
3      3 Charan   IT   90   88         95
4      4  Deepu  CSE   85   87         90
5      5  Eshan   IT   89   91         93
```

4.6:

```
> payroll <- rbind(payroll, new_emp)
> payroll
  EmpID  Name    Dept Basic  HRA   DA Rating
1   101 Rahul      IT 30000 6000 3000    4.5
2   102 Sneha      HR 35000 7000 3500    4.2
3   103 Kiran Finance 40000 8000 4000    4.8
>
```

4.7:

```
> hospital_data <- rbind(hospital_data, new_patient)
> hospital_data
  PatientID    Name Age      Disease Amount
1       201  Suresh  45     Diabetes  15000
2       202 Lakshmi  38 Hypertension  12000
3       203   Arjun  52      Cardiac  25000
>
```

4.8:

```
> sales_data <- rbind(sales_data, new_product)
> sales_data
  ProductID ProductName UnitsSold Revenue
1         1      Mobile       120 1200000
2         2      Laptop        80 4000000
3         3      Tablet        60 1800000
4         4 Smart Watch        90  900000
>
```

4.9:

```
> agri_data <- merge(agri_data, yield, by = "FieldID", all.x = TRUE)
> agri_data
  FieldID  Crop Moisture Temp ExpectedYield
1       1  Rice       55   32            NA
2       2 Wheat       47   29           3.5
3       3 Maize       NA   NA           4.1
```

4.10:

```
> final_records <- merge(student_data, attendance, by = "RollNo")
> final_records
  RollNo   Name Dept Sem1 Sem2 Attendance
1      1   Arun  CSE   78   80         92
2      2   Bina  ECE   82   85         88
3      3 Charan   IT   90   88         95
>
```

4.11:

```
> payroll <- merge(emp_details, salary, by = "EmpID", all.x = TRUE)
> payroll <- merge(payroll, rating, by = "EmpID", all.x = TRUE)
> payroll
  EmpID  Name    Dept Basic  HRA   DA Rating
1   101 Rahul      IT 30000 6000 3000    4.5
2   102 Sneha      HR 35000 7000 3500     NA
3   103 Kiran Finance    NA   NA   NA    4.8
```

4.12:

```
> # Merge with billing
> hospital_data <- merge(hospital_data, billing, by = "PatientID")
> hospital_data
  PatientID    Name Age  Disease Amount
1       201  Suresh  45 Diabetes  15000
2       202 Lakshmi  38     <NA>  12000
3       203   Arjun  52  Cardiac  25000
>
```

4.13:

```
> sales_data
  ProductID ProductName UnitsSold Revenue
1         1      Mobile       120 1200000
2         2      Laptop        80 4000000
3         3      Tablet        60      NA
4         4 Smart Watch        NA  900000
>
```

4.14:

```
> agri_data <- merge(agri_data, yield, by = "FieldID", all.x = TRUE)
> agri_data
  FieldID  Crop Moisture Temp ExpectedYield
1       1  Rice       55   32            NA
2       2 Wheat       47   29           3.5
3       3 Maize       NA   NA           4.1
>
```

5.1:

```
> names <- c("harshi", "priya", "sohan")
> upper_names <- toupper(names)
> name_length <- nchar(upper_names)
> upper_names
[1] "HARSHI" "PRIYA"  "SOHAN"
> name_length
[1] 6 5 5
```

5.2:

```
> emails <- c("arun@gmail.com", "harshi@yahoo.com", "santhu@outlook.com")
> usernames <- sub("@.*", "", emails)
> usernames
[1] "arun"   "harshi" "santhu"
```

5.3:

```
> input <- c(" R Programming ", " Data Science ", " AI ")
> clean_input <- trimws(input)
> clean_input
[1] "R Programming" "Data Science"  "AI"
```

5.4:

```
> product_codes <- c("WRD101", "WRD202", "ABC303")
> valid_codes <- startsWith(product_codes, "WRD")
> valid_codes
[1]  TRUE  TRUE FALSE
```

5.5:

```
> report <- "error in file, error in system, error occurred"
> updated_report <- gsub("error", "issue", report)
> updated_report
[1] "issue in file, issue in system, issue occurred"
```

5.6:

```
> password <- "harshi123"
> if(nchar(password) >= 9){
+   print("Strong Password")
+ } else {
+   print("Weak Password")
+ }
[1] "Strong Password"
```

5.7:

```
> sentence <- "R is a powerful programming language"
> word_count <- length(strsplit(sentence, " ")[[1]])
> word_count
[1] 6
```

5.8:

```
> files <- c("data.csv", "report.pdf", "image.png")
> extensions <- sub(".*\\.", "", files)
> extensions
[1] "csv" "pdf" "png"
```

5.9:

```
> names <- c("harshitha", "ramarao", "sridevi")
> abbr <- substr(names, 1, 3)
> abbr
[1] "har" "ram" "sri"
```

5.10:

```
> names <- c("Arun", "Bala", "Ravi", "John")
> matched_names <- grep("a", names, value = TRUE)
> matched_names
[1] "Bala" "Ravi"
```

5.11:

```
> clean_students <- students[students$ValidEmail == TRUE, ]
> clean_students
      Name         Email Department Username ValidEmail
1 GAYATHRI gayu@univ.edu        CSE     gayu       TRUE
2     RAVI ravi@univ.edu        ECE     ravi       TRUE
```

5.12

```
> products <- data.frame(
+   Product = c(" apple phone ", "SAMSUNG-TV ", " dell laptop"),
+   Code = c("PRD-101-APL", "PRD-202-SAM", "PRD-303-DEL")
+ )
> products$Product <- trimws(products$Product)
> products$Product <- tools::toTitleCase(tolower(products$Product))
> products$ProductCode <- sub("PRD-[0-9]+-", "", products$Code)
> products
       Product        Code ProductCode
1 Apple Phone PRD-101-APL          APL
2  Samsung-Tv PRD-202-SAM          SAM
3 Dell Laptop PRD-303-DEL          DEL
```

5.13:

```
> comments$CleanComment <- gsub("bad", "unacceptable", comments$Comment)
> comments
  User             Comment                 CleanComment
1    A      this app is bad     this app is unacceptable
2    B very bad experience very unacceptable experience
3    C       not bad at all      not unacceptable at all
```

6.1::

```
 marks <- c(55, 65, 75, 88, 95)
 total <- sum(marks)
 average <- mean(marks)
 highest <- max(marks)
 total
1] 378
 average
1] 75.6
 highest
1] 95
```

6.2:

```
> temperature <- c(10, 20, 25, 44, 25, 55, 33)
> min_temp <- min(temperature)
> max_temp <- max(temperature)
> min_temp
[1] 10
> max_temp
[1] 55
```

6.3:

```
> employee <- list(
+   ID = 315,
+   Name = "harshitha",
+   Salary = 45000,
+   Department = "IT"
+ )
> employee
$ID
[1] 315

$Name
[1] "harshitha"

$Salary
[1] 45000

$Department
[1] "IT"
```

6.4:

```
> patient
$PatientID
[1] 202

$Name
[1] "Raju"

$Age
[1] 55

$TestResults
[1] 120 140 130

> patient$TestResults
[1] 120 140 130
```

6.5:

```
> students <- data.frame(
+   RollNo = c(1, 2, 3),
+   Name = c("harshitha", "Karthik", "pranitha"),
+   Dept = c("CSE", "ECE", "IT"),
+   Marks = c(85, 78, 92)
+ )
> students
  RollNo      Name Dept Marks
1      1 harshitha  CSE    85
2      2   Karthik  ECE    78
3      3  pranitha   IT    92
>
```

6.6:

```
> sales <- data.frame(
+   Month = c("Jan", "Feb", "Mar"),
+   Sales = c(50000, 62000, 58000)
+ )
> total_sales <- sum(sales$Sales)
> total_sales
[1] 170000
>
```

6.7:

```
> prices
[1] 1200  850  560
> customer
$CustomerID
[1] 301

$Name
[1] "Anjali"

$City
[1] "Chennai"

> orders
     Item Price
1   Phone  1200
2 Headset   850
3 Charger   560
>
```

6.8:

```
> student_profile
$Name
[1] "Arun"

$RollNo
[1] 105

$Marks
[1] 78 85 88

> class_results
  RollNo Total
1    101   240
2    102   255
3    103   270
>
```

6.9:

```
> balances
[1] 5000 5200 5100
> customer
$AccountNo
[1] 954232

$Name
[1] "harshitha"

> transactions
   Date Amount
1 01-01   -600
2 02-01   7000
3 03-01   -600
>
```

6.10:

```
> # Display outputs
> print(profile)
$Name
[1] "Priya"

$Age
[1] 19

$Scores
[1] 50 60 70

> print(records)
  Subject Marks
1    Math    50
2      CS    60
3 Physics    70
>
```

7:

```
> write.csv(sales, "sales_data.csv", row.names = FALSE)
> # Read the CSV file
> sales <- read.csv("sales_data.csv")
> # View first few rows
> head(sales)
        Date Product    Category
1 2026-01-01   Phone Electronics
2 2026-01-01     Bag Accessories
3 2026-01-02  Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03     Bag Accessories
6 2026-01-03   Phone Electronics
  Quantity Price
1        5 20000
2       10  1500
3        3 50000
4        7  2500
5        5  1500
6        3 20000
> # Check structure and summary
> str(sales)
'data.frame':	6 obs. of  5 variables:
 $ Date    : chr  "2026-01-01" "2026-01-01" "2026-01-02" "2026-01-02"
...
 $ Product : chr  "Phone" "Bag" "Laptop" "Headset" ...
 $ Category: chr  "Electronics" "Accessories" "Electronics" "Electronic
s" ...
 $ Quantity: int  5 10 3 7 5 3
 $ Price   : int  20000 1500 50000 2500 1500 20000
> summary(sales)
```

```
> summary(sales)
     Date
 Length:6
 Class :character
 Mode  :character


   Product
 Length:6
 Class :character
 Mode  :character


   Category
 Length:6
 Class :character
 Mode  :character


    Quantity        Price
 Min.   : 3.0   Min.   : 1500
 1st Qu.: 3.5   1st Qu.: 1750
 Median : 5.0   Median :11250
 Mean   : 5.5   Mean   :15917
 3rd Qu.: 6.5   3rd Qu.:20000
 Max.   :10.0   Max.   :50000
> # View cleaned data
```

```
> head(sales)
        Date Product    Category
1 2026-01-01   Phone Electronics
2 2026-01-01     Bag Accessories
3 2026-01-02  Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03     Bag Accessories
6 2026-01-03   Phone Electronics
  Quantity Price
1        5 20000
2       10  1500
3        3 50000
4        7  2500
5        5  1500
6        3 20000
> # Remove extra spaces in Product and Category columns
> sales$Product <- trimws(sales$Product)
> sales$Category <- trimws(sales$Category)
> # Convert Date column to Date format
> sales$Date <- as.Date(sales$Date, format="%Y-%m-%d")
> # Add a column for total sales per row
> sales$TotalSales <- sales$Quantity * sales$Price
> # View cleaned data
> head(sales)
        Date Product    Category
1 2026-01-01   Phone Electronics
2 2026-01-01     Bag Accessories
3 2026-01-02  Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03     Bag Accessories
```

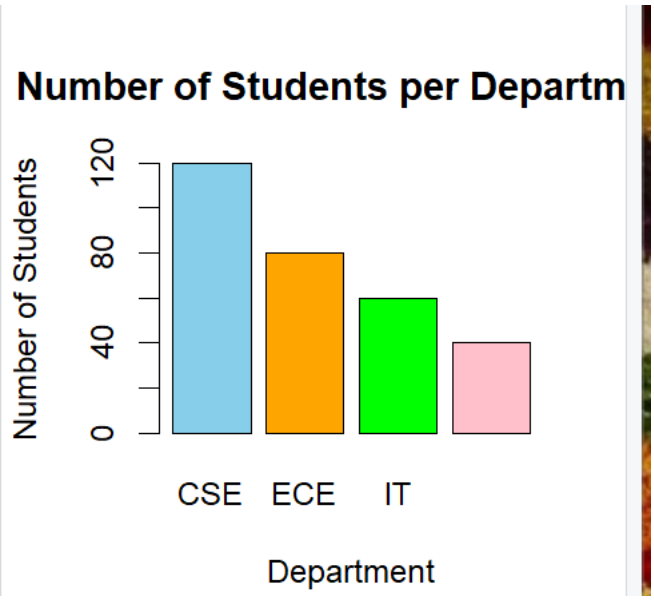
```
5 2026-01-03     Bag Accessories
6 2026-01-03   Phone Electronics
  Quantity Price TotalSales
1        5 20000     100000
2       10  1500      15000
3        3 50000     150000
4        7  2500      17500
5        5  1500       7500
6        3 20000      60000
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, su
m)
> total_sales_product
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
```

```
> # Calculate total sales for each product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, su
m)
> total_sales_product
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
>  Product with Highest Sales
```

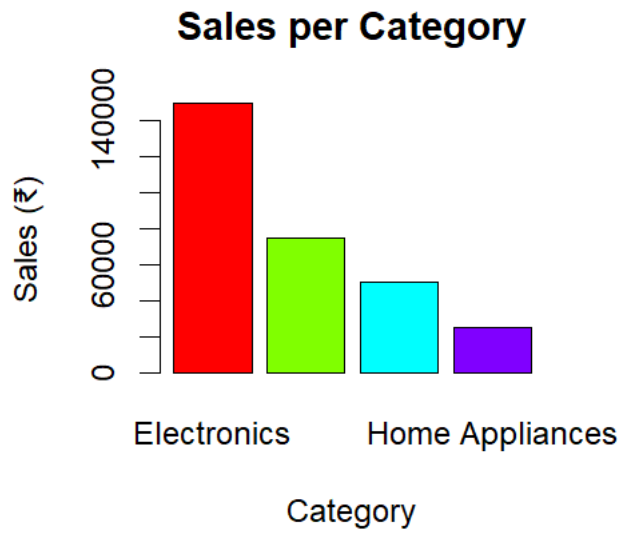
```
> top_product <- total_sales_product[which.max(total_sales_product$Tota
lSales), ]
> top_product
  Product TotalSales
4   Phone     160000
```

```
> quantity_per_category <- aggregate(Quantity ~ Category, data=sales, s
um)
> quantity_per_category
     Category Quantity
1 Accessories       15
2 Electronics       18
> high_sales <- total_sales_product[total_sales_product$TotalSales > 10
000, ]
> high_sales
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
>
```

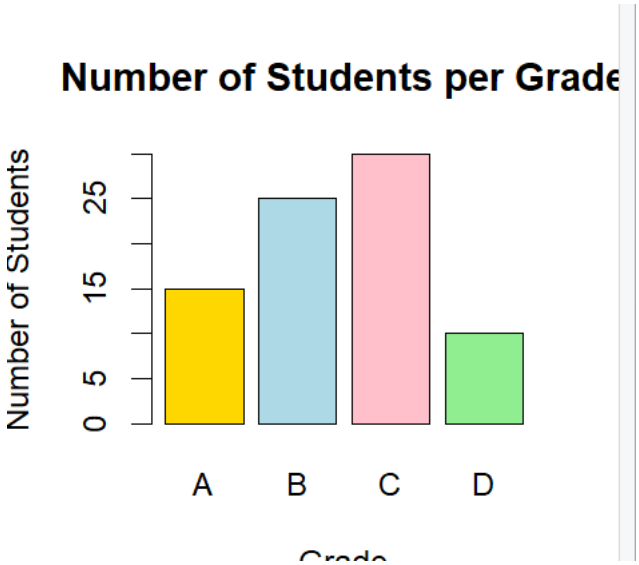
8.1:



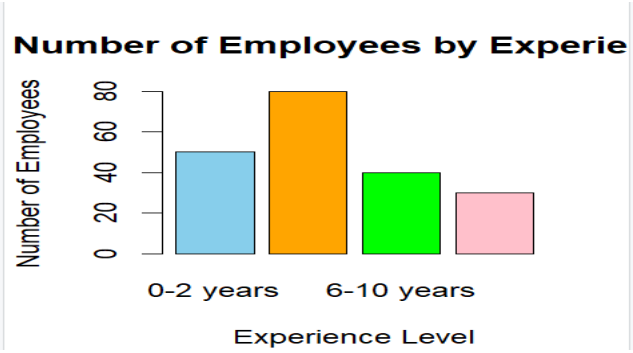
8.2:


Sales per Category
Sales (₹)
0
60000
140000
Electronics
Home Appliances
Category

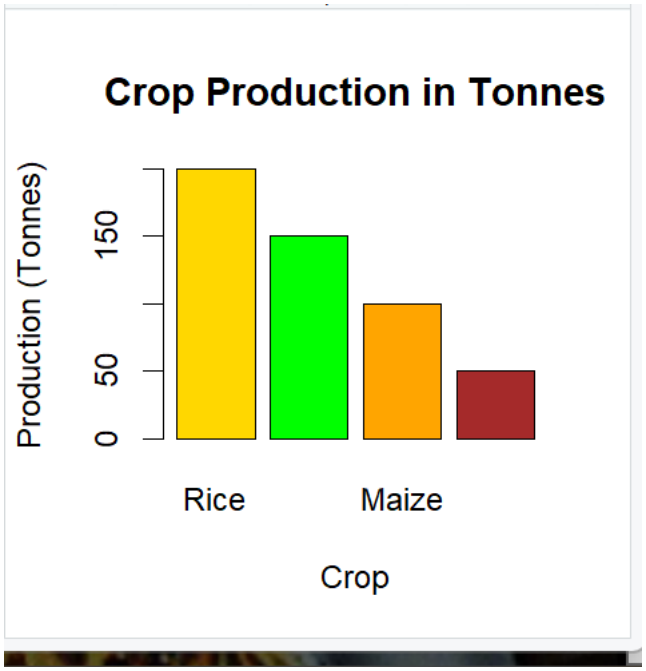
8.3:


Number of Students per Grade
Number of Students
0
5
15
25
A
B
C
D

8.4:


Number of Employees by Experie
Number of Employees
0
20
40
60
80
0-2 years
6-10 years
Experience Level

8.5:




9.1:

```
     Name Maths Science English
10 LAHARI    96      92      97
   Total
10   285
> above_avg_math
      Name Maths Science English
3  ROSHINI    90      80      85
4     ARYA    85      72      88
7     RAVI    85      70      88
9  SABITHA    98      95      70
10  LAHARI    96      92      97
   Total
3    255
4    245
7    243
9    263
10   285
>
```

9.2:

```
+ )
> # Step 3: Product with max aver
e sales
> max_avg_product <- sales_stats[
ich.max(sales_stats$Mean), ]
> sales_stats
   Product Mean Max Min Variance
1    Phone 27.4  47  14    194.3
2   Laptop 37.2  53  22    139.7
3  Headset 42.6  68  17    452.3
4      Bag 18.0  38  10    137.5
5   Camera 36.2  53  12    333.7
6    Mouse 31.6  39  18     85.3
7 Keyboard 28.4  36  25     19.3
> max_avg_product
  Product Mean Max Min Variance
3 Headset 42.6  68  17    452.3
>
```

9.3:

```
> older_patients <- patients[patien
ts$Age > 50, ]
>
> # Display results
> mean_age
[1] 45.4
> median_age
[1] 46.5
> range_age
[1] 29 60
> sd_age
[1] 9.547542
> older_patients
  PatientID   Name Age
3       203 ASHISH  52
5       205   RAVI  60
7       207 KALYAN  55
>
```

9.4:

```
> print(high_earners)
  EmpID    Name    Dept Salary
1   101    RAMU      IT 500000
2   102 SARANYA      HR 750000
6   106  GANESH Finance 742000
7   107    SIRI      IT 831000
>
```

9.5:

```
> # Display output
> mean_rating
[1] 6.3
> median_rating
[1] 6.5
> mode_rating
[1] 9
> var_rating
[1] 4.233333
> top_rated
   ProductID Product Rating
1          1   Phone      9
4          4     Bag      7
6          6   Mouse      6
8          8 Charger      8
9          9  Tablet      9
10        10 Speaker      7
>
```